

REPORT 5.1 - D1 Part A: where the hole is - build v33 (Opus ccae4e3) and Fable's review

2026-09-28. Spec: docs/SPEC_hole-position.md Sections 3 to 8. Built by Opus at xhigh from 1692724; one run, 16:03 to 16:35 CDT, 34 min and 486k sub-agent tokens by the harness count (n = 1). Filed by Fable from Opus's hand-back. Sections 1 to 7 are Opus's findings; Section 8 is Fable's review at xhigh.

1. What changed

File	Change
js/round/hole-position.js (new)	HOLE_ON_GREEN_M 15, PIN_OFF_GREEN_M 3, PIN_SHEET_UNCERTAINTY_M 4, APPROACH_BACK_M 137.16; greenFrame, pinFromSheet, offOwnGreenM, resolveHolePosition. Map order: cup (not track), pin sheet, ball on the green, green centre, with the filters and skipped. No map, or no green for the hole: the v32 order. Imports only geo.js and polygon.js.
js/round/round.js	holePosition(hole, context) returns what resolveHolePosition returns. New holeContextFor, setPinSheet, clearPinSheet. shotGeometry: a map tee reads hole.yards, source scorecard, lengthM null; a shot with no next mark ends at the hole only when the source is cup.
js/analysis/strokes-gained.js	context replaces fallbackPos; contextFor replaces fallbackFor; every shot carries distanceUncertaintyYd; the round result adds sources and positionNotes. Formula, categories and benchmarks untouched.
js/analysis/trends.js	Both call sites use holeContextFor.
js/data/schema.js	newHole gets pinSheet: null and its comment. Nothing else.
js/ui/screen-summary.js	The sources line, one line per position passed over, the 5.5 yd (n = 31) line; the drive card and the data quality card read the resolved position.
js/ui/screen-play.js	shotRowHeading(row, i, { pin, yards }) with the source suffix; the shot list's second line; the map available in edit mode, the HUD still off there.
sw.js, js/data/build.js	v33; hole-position.js in the shell list. REVISION untouched.
test/run.js	The 13 tests; 6 existing tests updated, none retired.

2. The spec's verify items

Item	Answer
3.4 no import cycle	34 modules, 0 cycles.
5 a round with pinSheet survives export then import	Yes, byte-identical, and resolves to the same pin (test 11).

3. The 13 tests

All 13 fail against the modules at 1692724 (a throw-away harness, deleted). Test 10 is a regression guard for a course with no map, its values computed at 1692724.

#	What it proved
1	No cup: 3 off-green shots, all map-green, each equal to toGreen().centreM within 1e-9 yd
2	A cup 6 yd from the centre wins; skipped empty
3	A cup from the track on the green is not used; one cup / from-track entry; the stored cup is still there
4	A cup 85 m off its green is not used; the same cup 5 to 15 m off is used
5	12 on, L 5, from the edge, 3 ft: 12 paces from the front along the line and 5 in from the left edge, within 0.2 m
6	C is halfway between the edges (0.05 m); centre R 4 is 4 paces right of the line
7	A pin sheet 5 paces deeper than the green falls to map-green, one pin-sheet entry, his numbers kept
8	Map tee on hole 15: 386 yd, scorecard, no length; a GPS tee unchanged
9	Map tee, two track shots, putt typed: the off-green sum is equal to 1e-9 with no pin, a front pin and a back pin; approach and short game each differ by more than 1e-3
10	Radcliffe: positions to 1e-12, strokes gained to 1e-9, equal to 1692724

#	What it proved
11	A missing key resolves like null; newHole gives null; the export round trip
12	Hole 9's frame matches a hand construction to 0.01 m; see Section 6 item 1
13	sources adds up to the 9 shots that are not putts; positionNotes names hole 1

Suite: 594/594 at 360x728 (580 before, plus 13, plus 1 offline-shell test for the new module).

4. Before and after, every Veenker round in the phone's store

Read only; nothing written. 16 rounds, 9 score at least one hole. Scratch baseline. "Before" is 1692724.

Round	Holes	Unattributed	Off tee	Approach	Short game	Putting	Total	Sources
09-09 21:41	1	1	-0.225	-0.359	-0.024	0.000	-0.608	cup 3, same
09-09 22:09	5	2	-0.461	-2.603	-0.708	-0.009	-3.781	cup 17, same
09-11	3	4	+0.064	-0.950	-1.405	+0.627	-1.664	cup 8, same
09-14	9	2	-1.657	-1.980	-1.199	-0.063	-4.898	cup 23, same
09-16	9	5	-1.833	-4.763	+0.571	+0.230	-5.795	cup 26, unknown 2, same
09-23	3	2	+0.977	0.000	0.000	-0.504	+0.473	cup 2, same
09-26 14:51 (abandoned)	1	0	-1.000 to -1.001	0.000 to +8.043	+0.170 to 0.000	-0.947	-1.778 to +6.094	cup 2 to map-green 2
09-26 17:48	1	0	-0.583	-2.788	+1.727	0.000	-1.644	cup 6, same
09-27	8	9 to 7	-3.700 to -1.750	+0.331 to +5.122	+1.936 to -0.598	+0.678	-0.755 to +3.452	cup 14, unknown 2 to map-green 16

- 7 of 9 rounds are unchanged to 3 decimals: every cup in them is a burst cup within 15 m of its green.
- 09-26 14:51: every mark on this abandoned round is about 3,894 yd from hole 1's green. Both the before and the after numbers are meaningless.
- 09-27: the v30 track picks, now measured to the right hole. His word on that round: "I am not worried about yesterdays round."
- Trends take completed rounds of 9 or more holes: 09-14 and 09-16 only, both unchanged.

5. The round trip and the frame

- 31 burst cups on their green, each described in whole paces from the nearer edge and placed back: median 0.36 m, largest 0.50 m from the cup (n = 31). Rounding to whole paces allows up to 0.65 m. This checks the arithmetic, not a real pin sheet.
- The frame, 18 holes: A is 150 yd back along the line on all 18. Depth 17.5 yd (hole 11) to 42.0 yd (hole 17); half depth, the error bar on the green centre, 8.8 to 21.0 yd.

6. Where the spec was wrong or silent

1. **Test 12's example was wrong.** Hole 11's line is 152.3 yd, longer than 150, so A is 2.1 m from its start, not at it. No Veenker line is shorter than 150 yd. Opus built Section 4.1 as written and proved the short-line branch on a trimmed copy.
2. **6.1 has no map condition**, so a source: 'map' tee on a course with no map reads the card instead of being unattributed.
3. A distance he typed on a map tee still beats the scorecard.
4. The shot list's second line replaces the shot's length for the scorecard, the pin sheet and the green centre: both together ran off the row at 360 px.
5. positionNotes entries carry hole and used; greenFrame returns approach.
6. ON 0 sits exactly on the front edge and may not place.

7. Follow-ups Opus noticed, not fixed

- A shot mark nowhere near its hole gives a lookup far off the table (09-26 14:51: +8.043 on one shot).
- endsAtCup still reads hole.cup of any method; no caller uses it.
- test/README.md has no D1 row.

8. Fable's review (xhigh), 2026-09-28

Check	Result
hole-position.js, round.js, strokes-gained.js, trends.js, schema.js	Read line by line against Sections 3 to 6.
screen-summary.js, screen-play.js	Read on code lines.
Must-not-touch list (Section 10)	10 files changed, none on the list. REVISION untouched. v33 in both files.
Suite	594/594 at 360x728, Fable's run, n = 1.
The pin arithmetic, independently	Fable's own construction (sampling at 0.01 m) against pinFromSheet on holes 1, 17, 9, 11 and 3: 0.002 to 0.011 m apart (n = 5).
The "after" column of Section 4	Recomputed by Fable from the phone's store with the built modules: all 9 rounds equal to 3 decimals. The "before" column is Opus's measurement.
The strokes gained card on screen	Rendered at 360 px for the 2026-09-27 round: the lines fit, no sideways scroll.
Commit identity	rusty9645@gmail.com.

Verdict: PASS against the spec, with three corrections owed before it goes anywhere (spec Section 12).

#	Correction	Why
C3	A map tee has no lengthM even when he typed its distance.	shotGeometry takes the typed branch first and measures a length from the box centre. 6.1 says a map tee has no length.
C4	The card's sentence "The total does not depend on it" shows only when no hole's first shot was measured to the green centre.	On the 2026-09-27 round the tees are track picks measured to the green centre, so the total there does depend on it. The sentence was false on the first round it was shown for. Fable's spec wrote it without the condition.
C8	A shot mark farther from the hole than the card yardage plus 100 yd is not on the hole and gives no distance.	Opus's follow-up 1. The engine "refuses to guess a distance"; a lookup 3,905 yd out is a guess.

Rulings on Opus's three calls: test 12's example is corrected in the spec (C1); 6.1 stands on every course, because the scorecard exists on every course (C2); follow-up 1 is built as C8.

Not installed, not pushed. By his word the deploy belongs to the Hole Overview session.